



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 9 • STANDARD 6.AT.11

# Write Equations to Represent Relationships Between Two Variables

Lesson 9-3 • Teacher Copy

**Name:** Type your name

**Date:** Today's date

**Class:** Class period

## My Notes

Level 2 Standard



### TODAY'S OBJECTIVES

**Content Objective:** I can write an equation from a table or a graph to model the relationship between the dependent and independent variables.

**Language Objective:** I can define the variables in my equation and explain what each part represents using the words equation, variable, and rate.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

equation

variable

rate

represent

define the variables



Tap any word to see what it means and a picture.

1 A mathematical sentence with an equals sign is an .

2 A letter standing for a quantity that can change is a .

3 A fixed amount of one quantity for every one unit of the other is a .

4 To show a relationship with a table, graph, or equation is to  it.

5 To state what each letter stands for is to   
 .

## Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

EXPLAIN

**Would you recommend that Diamond purchase or stream songs from the online music service?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

## 2. Plan Your Math Words

Check at least two words you will use.

☐ **equation** — A mathematical sentence with an equals sign that shows two amounts are the same.  
*ecuación — Una oración matemática con un signo igual que muestra que dos cantidades son iguales.*

☐ **variable** — A letter that represents a quantity that can change, or vary.  
*variable — Una letra que representa una cantidad que puede cambiar o variar.*

☐ **rate** — A fixed amount of one quantity for every one unit of the other quantity.  
*tasa — Una cantidad fija de una cantidad por cada unidad de la otra cantidad.*

☐ **represent** — To show a relationship using a tool such as a table, a graph, or an equation.  
*representar — Mostrar una relación usando una herramienta como una tabla, una gráfica o una ecuación.*

☐ **define the variables** — To state exactly what quantity each letter in an equation stands for.  
*definir las variables — Decir exactamente qué cantidad representa cada letra de una ecuación.*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**Diamond should write an \_\_\_\_ for each option so she can \_\_\_\_ the costs.**  
**Each equation uses the \_\_\_\_ as the input and gives the \_\_\_\_ as the output.**

*Di tu respuesta primero: Mi respuesta es \_\_\_\_ porque \_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** The fee is charged EVERY month, so the total must be the rate multiplied by the number of months, not added to it once.

**Evidence:** A strong answer tests the equation against a table row (2 months should give \$49.90) rather than just stating a rule. A weak answer picks multiplication without checking it against the actual numbers.

**Reasoning:** This evidence connects the definition of equation to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- **Diamond should write an \_\_\_\_ for each option so she can \_\_\_\_ the costs. Each equation uses the \_\_\_\_ as the input and gives the \_\_\_\_ as the output.**
- **My answer is \_\_\_\_.**

*Mi respuesta es \_\_\_\_.*

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **I know because \_\_\_\_.**  
*Lo sé porque \_\_\_\_.*
- **So \_\_\_\_.**  
*Entonces \_\_\_\_.*

## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*
- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*
- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*
- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

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## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

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## Answer Key & Teacher Guide

1. **Notes 1:** equation
2. **Notes 2:** variable
3. **Notes 3:** rate
4. **Notes 4:** represent
5. **Notes 5:** define the variables
6. **Watch for this mistake:** *Writing the equation with addition —  $c = 24.95 + m$  — instead of multiplication. The fee is charged EVERY month, so the total is the rate times the number of months; always test the equation against a row of the table.*
7. **Try It 1:** A. The equation gives the total cost for ANY number of months, not just the rows listed — *A table can only show the values someone wrote down; substituting any value of  $m$  into  $c = 24.95m$  gives the cost for that many months instantly.*
8. **Try It 2:** A. 8 hours — *Divide both sides by 15:  $h = 120 \div 15 = 8$ .*